

New method for radar HRRP recognition and rejection based on weighted majority voting combination of multiple classifiers

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Abstract—To deal with the problems of recognition and outlier rejection in radar automatic target recognition (RATR), a novel method based on weighted majority voting combination of multiple classifiers is proposed. In both the rejection and recognition phases, the results of multiple classifiers which employ different features are combined. In the rejection part, the nearest neighbor (NN) classifier with the outputs of a classifier as the input features is utilized to identify outliers. One cost function is proposed to measure the recognition and rejection performance of each classifier. Experimental results based on the measured HRRP dataset validate the effectiveness of the proposed method.

Keywords—recognition; rejection; RATR; weighted majority voting; combination of multiple classifiers; cost function

I. INTRODUCTION

Radar high resolution range profile (HRRP) contains lots of target structure information, so that it has strong ability for target identification. It also has such advantages as easy acquisition, small storage needed and so on. Because of these, target recognition based on radar HRRP has received broad attention in the radar automatic target recognition (RATR) community [1–3].

In RATR community, most of the targets are uncooperative or even hostile. In that case it will be difficult or unrealizable to get complete training database. When a target not belonging to any class of the database (outlier) is detected, it is unreasonable to recognize it as a target in the database (inner) [4]. So the recognition rate cannot be the only rule to evaluate the performance of a classifier here, the ability to reject outliers should also be taken into account. In Section 5.1 a cost function is proposed to measure the recognition and rejection performance of a classifier.

For rejection problem, we deal with a two-class classification problem. One is the target class (or named as inner), the other is the outlier class. However, the outlier class might be very expensive to be measured [5–7]. Generally there are two methods to solve this problem: (1) using one –class classification [5–7], (2) generating outliers [8]. Using the method proposed in [8], an outlier database is generated, which is used to train the classifier and test its performance.

The problem of classifier combination has always been of interest to the pattern recognition community. In the recognition process, different features or classifiers may lead to different recognition results. To a certain extent, they are complementary. In [9–12], classifier combination rules are implemented to improve the recognition performances of individual classifiers. In the same way, we can also improve the rejection performance of individual classifiers by combining multiple classifiers. Therefore in this paper we combine the powers of multiple classifiers in the both parts of rejection and recognition, to improve both the recognition and rejection performance.

II. FEATURE EXTRACTION AND BRIEF INTRODUCTION TO CLASSIFIERS

By adopting combination rules, targets can be effectively classified by some simple classifiers via employing a small set of inexpensive features, thereby reducing the complexity of the system. In this paper, we use three simple classifiers each of which employs an ordinary feature: (1) Maximum correlation classifier (MCC [2]) employing real HRRP, (2) Relevant vector machine (RVM [13]) employing the amplitude of frequency spectrum, (3) Support vector machine (SVM [14]) employing power spectrum. Let $X(t)$ be a real and continuous function of time t .

1) The amplitude of frequency spectrum

$$A(\omega) = |\text{FT}(X(t))| \quad (1)$$

2) The power spectrum

$$P(\omega) = |\text{FT}(X(t))|^2 \quad (2)$$

where $|\cdot|$ denotes modulus operation, $\text{FT}(\cdot)$ denotes Fourier transform.

III. THE NN CLASSIFIER USED FOR REJECTION

Before the introduction of the NN classifier, we talk about the method of generating outlier examples artificially. It is a reasonable assumption that the outliers are uniformly distributed in the full sample space. However, because of the limitation of the number of generated outliers, it is difficult to generate full-space uniformly distributed examples, especially

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for the high-dimensional HRRP data. So it is necessary to limit the range of the distribution of each dimension of the outliers. Here we adopt the method 2 proposed in [8], only enlarging the range by 2 times. Take 2-dimensional inners for example. Suppose their first dimension is $[-1, 1]$ uniformly distributed and the second dimension is $[-10, 10]$ uniformly distributed. Then we suppose the range of the first dimension of outliers is $[-2, 2]$, while the second dimension $[-20, 20]$.

The nearest neighbor (NN) classifier is a popular tool in the pattern recognition community. Here it is used to deal with outlier rejection problems as follows.

- a) Given a testing sample $\mathbf{x}$, compute the Euclidean distance between it and each training data.
- b) Find its nearest neighbor $\mathbf{x}_{NN}$ and calculate the distance between them $d_{\mathbf{x}, \mathbf{x}_{NN}} = \|\mathbf{x} - \mathbf{x}_{NN}\|_2$.
- c) Compare the distance $d_{\mathbf{x}, \mathbf{x}_{NN}}$ with the threshold t which is calculated in the training step. If $d_{\mathbf{x}, \mathbf{x}_{NN}} < t$, then $\mathbf{x}$ is an inner, otherwise $\mathbf{x}$ is an outlier.

However, when the NN classifier is adopted, the decision is only relative to the nearest neighbor of a testing example. In order to derive more information from the training data, we propose a new method to take the output of the classifier, which is used to classify the testing example, as the feature employed by NN the classifier. Fig.1 shows the process of recognition and rejection. The output of the system contains two parts of information: one is whether the sample is an inner or not and the other is the label of the class the sample belongs to.



Figure 1. The process of recognition and rejection.

A classifier can act as a feature extractor [15], i.e. the output of a classifier can be taken as a feature of the input sample. The output of a classifier reflects the relation between the input sample and the training samples of each class, so it contains more information. The dimension of the output equals to the number of target class, and then the dimension of the feature may decrease. In this paper, the number of class is 3 and the number of the feature extracted is no less than 128. Therefore using the output of a class reduces the computing complexity.

IV. WEIGHTED MAJORITY VOTING PRINCIPLE

The weighted majority voting principle is a simple and practical method of classifier combination [9, 10]. Compared with the majority voting method, it has each classifier assigned a weight, which is associated with its performance. The common method is to calculate the weight of a classifier with its confusion matrix. The detailed process is shown as follows.

We make an assumption that there are M classes $C_1, C_2, \dots, C_M$ in the database, with each of C_i denoting the i th class. There are K classifiers $e_1, e_2, \dots, e_K$, where

each e_k denotes the k th classifier. $e_k(\mathbf{x}) = j, k = 1, 2, \dots, K; j \in 1, 2, \dots, M$ denotes that the classifier e_k classify the sample $\mathbf{x}$ as the j th class. The confusion matrix of each classifier e_k is given by (3).

$$PT_k = \begin{bmatrix} n_{11}^{(k)} & n_{12}^{(k)} & \dots & n_{1M}^{(k)} \\ n_{21}^{(k)} & n_{22}^{(k)} & \dots & n_{2M}^{(k)} \\ \vdots & \vdots & \ddots & \vdots \\ n_{M1}^{(k)} & n_{M2}^{(k)} & \dots & n_{MM}^{(k)} \end{bmatrix}, k = 1, 2, \dots, K \quad (3)$$

where $n_{ij}^{(k)}$ denotes that $n_{ij}^{(k)}$ samples of class C_i have been classified as the j th class by e_k .

The confusion matrix PT_k can be obtained by using e_k to classify a testing sample set. For classifier e_k , with its knowledge of the confusion matrix PT_k , the probabilities that propositions $\mathbf{x} \in C_i, i = 1, 2, \dots, M$ are true under the occurrence of the event $e_k(\mathbf{x}) = j$ is give by (4).

$$p_{ij}^k = p(\mathbf{x} \in C_i | e_k(\mathbf{x}) = j) = n_{ij}^{(k)} / \sum_{i=1}^M n_{ij}^{(k)}, i = 1, \dots, M \quad (4)$$

Then we can get the weight matrix of each classifier as follows:

$$W_k = \begin{bmatrix} p_{11}^{(k)} & p_{12}^{(k)} & \dots & p_{1M}^{(k)} \\ p_{21}^{(k)} & p_{22}^{(k)} & \dots & p_{2M}^{(k)} \\ \vdots & \vdots & \ddots & \vdots \\ p_{M1}^{(k)} & p_{M2}^{(k)} & \dots & p_{MM}^{(k)} \end{bmatrix}, k = 1, 2, \dots, K \quad (5)$$

So the final decision made by the combined classifier E is given by:

$$E(\mathbf{x}) = \arg \max_i \sum_{k=1}^K p_{ij_k}^k, i = 1, 2, \dots, M; \quad (6)$$

where $j_k \in \{1, 2, \dots, M\}$ denotes the label that classifier e_k assigns to sample $\mathbf{x}$.

V. RADAR HRRP REJECTION AND RECOGNITION BASED ON CLASSIFIER COMBINATION

A. Cost Function

The rejection performance of a classifier is generally evaluated by the Receiver Operating Characteristic (ROC) curve. The two parameters of the curve are the false positive rate P_f and the true positive rate P_d , which are respectively defined as (7) and (8). To measure the recognition performance, the recognition rate P_a is defined as the average recognition rate of the samples classified as inner ones.

$$P_f = \frac{\text{number of inner samples classified as inner ones}}{\text{number of all inner samples}} \quad (7)$$

$$P_d = \frac{\text{number of outlier samples classified as inner ones}}{\text{number of all outlier samples}} \quad (8)$$

There are three kinds of erroneous cases in the output of the classifier system: 1) classify inner samples as outlier ones, 2) falsely classify the inner samples that have been classified as inner ones, 3) classify outlier samples as inner ones. Supposing that error 1) and error2) have the same cost, we propose a cost function as:

$$CF = c_1(1 - P_d P_a) + c_2 P_f \quad (9)$$

where c_1 denotes the cost of error 1) and 2), c_2 denotes the cost of error 3).

For simplicity, we define the cost ratio $\alpha = c_2 / c_1$, and then the cost function becomes(10).

$$CF = \frac{1}{1 + \alpha} ((1 - P_d P_a) + \alpha P_f) \quad (10)$$

where the constant $\frac{1}{1 + \alpha}$ is used to normalize the cost function.

B. Rejection and recognition

We divide the database into three data sets: training set 1 which contains inner samples only; training set 2 which contains both inner and outlier samples; testing set which contains both inner and outlier samples. The whole system is consisted of two stages.

1) Training Stage

The training process is consisted of two steps: step1, use training set1 to train the three classifiers in Section 2; step2, for each classifier, find the threshold t and calculate the rejection and recognition weights of each classifier. The details of step2 are as follows:

- Classify the samples in training set 2.*
- Adopting the NN classifier which employs the outputs as features, we can get the nearest distance set $\{d_i\}, i=1,2,\dots,N$, where N denotes the number of samples in training set 2.*
- Search the threshold t : let $t = d_i, i=1,2,\dots,N$, calculate the value of cost function $CF_i, i=1,2,\dots,N$, finally let $t = d_j$, if $CF_j = \min_i CF_i$.*
- Compute the rejection confusion matrix and the recognition confusion matrix, then calculate the rejection and recognition weights of each classifier.*
- Record the threshold and the weight matrices.*

2) Testing Stage

We use the testing set to test the whole system. For a testing sample $\mathbf{x}$, we adopt testing steps as follows.

- Extract the features of the sample: real HRRP, the amplitude of frequency spectrum and the power spectrum.*
- Use the three classifiers MCC, RVM and SVM to classify the sample.*
- Implement rejection process using NN classifiers.*
- Classify the sample as an inner one or not, based on weighted majority voting method.*
- If the sample is classified as an inner one, assign its label by the combined classifier E, otherwise output the sample as an outlier sample.*

The testing process is shown in Fig. 2. In Fig. 2, RHRRP denotes real HRRP feature; AFS denotes the amplitude of frequency spectrum feature; PS denotes power spectrum feature; CC1 denotes combining the result of rejection; CC2 denotes the result of recognition. From the process, we can see that the proposed algorithm can allocate task to several processing units to implement parallel processing and increase the efficiency.

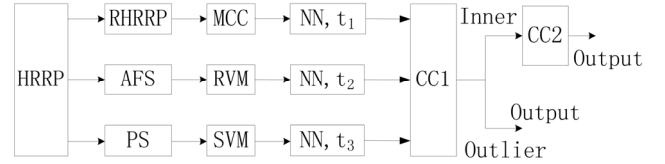


Figure 2. The testing process.

VI. EXPERIMENTS

We use the measured ISAR (Inverse Synthetic Aperture Radar) data of 3 classes of airplane as inner examples, and then artificially generate outlier examples to conduct experiments. The parameters of radar and airplanes are shown in Table 1. The inner contains three targets: Yark-42, AN-26 and Cessna Citation S/II. The training set 1, training set 2 and testing set separately contains 90000 HRRPs, 30000 HRRPs and 5200 HRRPs. 50000 outliers are generated artificially, 30000 outliers for training set 2 and the other 20000 outliers for testing set.

TABLE I. PARAMETERS OF PLANES AND RADAR IN THE ISAR EXPERIMENT

radar parameters	center frequency		5520 MHz
	bandwidth		400 MHz
planes	length(m)	width(m)	height(m)
Yark-42	36.38	34.88	9.83
An-26	23.80	29.20	9.83
Cessna Citation S/II	14.40	15.90	4.57

In the experiment, the value of cost ratio α varies from 0.5 to 2 with the increment of 0.1. Based on the cost function, we compare the performance of the individual classifiers (MCC, RVM, SVM) and combined classifier (CC). With different cost ratio, the values of the parameters of the cost function, P_d , P_f and P_a , are shown in Fig. 3. The values of cost function are shown in Fig. 4.

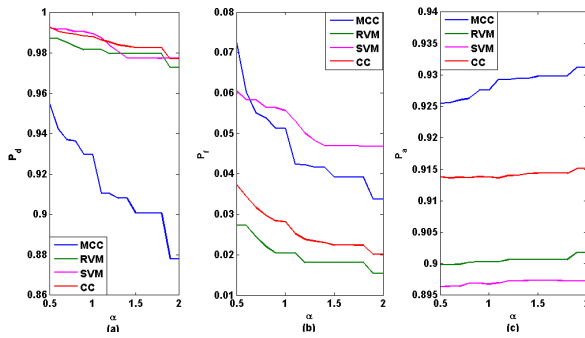


Figure 3. The values of the parameters versus α . (a) P_d versus α . (b) P_f versus α . (c) P_a versus α .

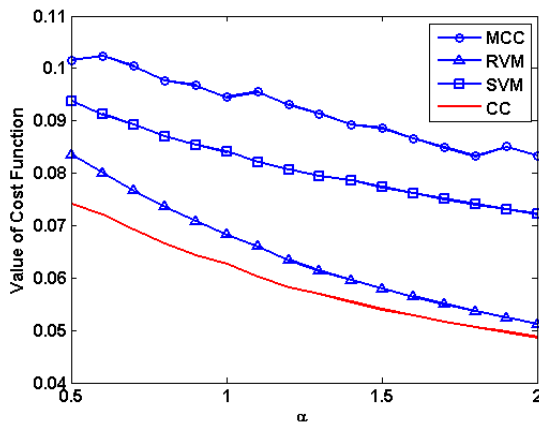


Figure 4. The cost function values of individual classifiers and the combined classifier.

The results of Fig. 3 show that different classifiers have different performance of recognition and rejection. Through the results shown in Fig. 4, we see that the cost function value of the combined classifier is lower than that of each individual classifier. The combined classifier improves the integrated performance. From the calculation results, we can see that the cost function value of the combined classifier decreases about 5.1-11.1% with respect to the optimal cost function value of single classifier.

VII. CONCLUSIONS

Based on weighted majority voting combination of multiple classifiers algorithm, we propose a method to combine both the recognition and rejection powers of several classifiers. On the basis of the proposed cost function, simulation results verify the advantage of the combined classifier over individual classifiers. Furthermore, the proposed algorithm can allocate task to several processing units to implement parallel processing and increase the efficiency. So it is an effective algorithm that can improve the performance of rejection and recognition in RATR.

REFERENCES

- [1] Shi L, Wang P, Liu H, Xu L, Bao Z. Radar HRRP statistical recognition with local factor analysis by automatic Bayesian Ying-Yang harmony learning. *IEEE Transactions on Signal Processing*, 2011, 59(2), pp. 610–617.
- [2] Wang, P. and Shi, L. and Du, L. and Liu, H. and Xu, L. and Bao, Z.. Radar HRRP statistical recognition with temporal factor analysis by automatic Bayesian Ying-Yang harmony learning. *Frontiers of Electrical and Electronic Engineering in China*, 2011, 6(2), pp.300-317.
- [3] Du Lan, Liu Hongwei, Bao Zheng, Zhang junying. A two-distribution compounded statistical model for radar HRRP target recognition. *IEEE Trans. on Signal Processing*, 2006, 54(6), pp. 2226-2238.
- [4] T.C.W. Landgrebe, D.M.J. Tax, P. Paclik, R.P.W. Duin, C. Andrew. "A combining strategy for ill-defined problems", Fifteenth Annual Symposium of the Pattern Recognition Association of South Africa, November, 2004.
- [5] Tax D, Duin R Support Vector Domain Description. *Pattern Recognition Letters*, 1999(20), pp. 1191-1199.
- [6] Tax D.M.J.. One-class classification. PhD Thesis, Delft Univ. of Technology, The Netherland, 2001.
- [7] PAN Zhi-song, CHEN Bin, MIAO Zhi-min, NI Gui-qiang. Overview of Study on One-Class Classifiers. *ACTA ELECTRONICA SINICA*, Nov. 2009, Vol.37(11), pp. 2496-2503.
- [8] CHAI Jing, LIU Hong-wei, BAO Zheng. Application of a weighted KNN classifier for HRRP out-of-database target rejection. *Systems Engineering and Electronics*, April 2010, Vol.32(4), pp. 718-723.
- [9] L. Xu, A. Krzyzak and C.Y. Suen, Several methods for combining multiple classifiers and their applications in handwritten character recognition, *IEEE Trans. on System, Man and Cybernetics SMC-22(3)* (1992), pp. 418–435.
- [10] J. Sun, H. Li. Listed companies' financial distress prediction based on weighted majority voting combination of multiple classifiers. *Expert Systems with Applications*, 2008, Vol. 35(3), pp. 818-827.
- [11] S. Prasad, L. M. Bruce. Decision fusion with confidence-based weight assignment for hyperspectral target recognition. *IEEE Trans. on Geoscience and Remote Sensing*, 2008, Vol. 46(6), pp 1448-1455.
- [12] CAO Xiang-hai, LIU Hong-wei, WU Shun-jun. Utilization of multiple polarization data and multiple features for radar target identification. *Systems Engineering and Electronics*, Feb. 2008, Vol.30(2), pp. 261-264.
- [13] Carin L, Dobeck G. Relevance Vector machine feature selection and classification for underwater targets. In: *proceedings of Oceans Conference Record (IEEE)*, 2003, pp. 22-26.
- [14] Duda R. O., Hart P. E., Stork D. G.. *Pattern Classification*. Second Edition. New York: John Wiley and Sons, 2001.
- [15] F. F. Soulie, E. Vinnet, and B. Larny, Multi-modular neural network architectures: Applications in optical character and human face recognition, *Int. J. Pattern Recognition and Artificial Intelligence*, 5, 1993(4), pp. 721-755.